Counterdiabatic Hamiltonian Monte Carlo

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Abstract

Hamiltonian Monte Carlo (HMC) is a state of the art method for sampling from distributions with differentiable densities, but struggles to converge to the stationary distribution for challenging multimodal problems. Running HMC with a time varying Hamiltonian, to interpolate from a tractable distribution to the target of interest results in bias, which can be corrected by weighting schemes, as in Sequential Monte Carlo sampling [**doucet2001introduction**]. However, this can be inefficient, since it requires slow change between the two distributions. Inspired by [**sels2017minimizing**], where a learned *counterdiabatic* term allows for efficient quantum state preparation, we propose *Counterdiabatic Hamiltonian Monte Carlo*. We establish its relationship to recent proposals for accelerating gradient-based sampling with learned drift terms, and show initial promising results.

1 Introduction

Algorithms for sampling from probability distributions have many application in the physical sciences [**tuckerman2023statistical**, **von2011bayesian**] and Bayesian statistics [**gamerman2006markov**]. For arbitrary distributions with differentiable density functions, *Hamiltonian Monte Carlo* (HMC) [**betancourt2017conceptual**] is a state of the art method for sampling, and in conjunction with the No-U-Turn scheme for trajectory length [**hoffman2014no**], it is a reliable and widely used method [**gelman2015stan**].

HMC is founded on a correspondence between probabilistic inference and classical statistical physics, and produces a Markov kernel based on a numerical simulation of Hamiltonian dynamics. This entails an interpretation of the stationary distribution (in the extended space of position and momentum) as the canonical equilibrium distribution of statistical mechanics, and interpretations of the logarithm of the density to the potential energy, force to the gradient of this logarithm, and reversibility to work [**sivak2013using**].

MCMC works by constructing a Markov kernel which has as its stationary distribution the target distribution, so that a Markov chain samples from the target distribution in the limit of many steps. However, for certain distributions, the number of steps of the chain required to reach the target distribution is prohibitively large. A motivating example is a mixture of two Gaussians with means separated by a distance much larger than the respective variances. If the chain is initialized near one of the two modes, attaining stationarity requires moving through areas of low probability density.

One proposed solution to this problem is a Sequential Monte Carlo (SMC) sampler [**del2006sequential**], which constructs a series of distributions, starting with an easy distribution (such as a Gaussian) and ending with the target (such as the Gaussian mixture), using a Markov kernel at each step in the series and a weighting scheme to remove bias. The challenge of SMC is to make each distribution in the series similar enough to the previous that the output of the Markov kernel does not lag far behind the current distribution (which results in a high variance weighting scheme), while making them different enough that only a small number of steps is required.

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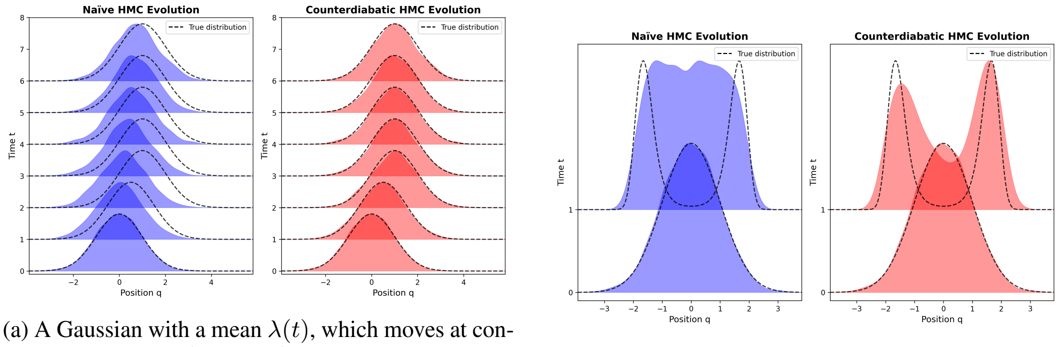

(a) A Gaussian with a mean $\lambda(t)$, which moves at constant velocity from $\lambda = 0$ to $\lambda = 1$, and then remains at 1. Simply running HMC at each intermediate time step (blue) results in a lag, which is accounted for by the counterdiabatic driving term (red).

(b) An interpolation between a Gaussian at $t = 1$ and a double well at $t = 1$. The counterdiabatic driving does not result in a perfect transport, but clearly accelerates the process.

Figure 1: Illustration of counterdiabatic driving in Hamiltonian Monte Carlo. Left: Moving mean example showing lag and correction. Right: Double well example showing accelerated transport.

In the setting of HMC, moving through a series of distributions in this way corresponds to time evolution of a physical system while changing the system's parameters, a central topic of thermodynamics and statistical mechanics. From this perspective, varying the parameters too quickly results in *lag*, or more physically *dissipation* [**vaikuntanathan2011escorted**], where the Hamiltonian flow cannot return to equilibrium quickly enough as the system's parameters change.

Several papers [**demirplak2003adiabatic**, **berry2009transitionless**], both in the quantum and classical setting, have noted that one can compensate for this lag by adding a *counterdiabatic driving term* to the Hamiltonian. More recently, a scheme to approximate this term variationally has been proposed in a quantum setting [**sels2017minimizing**]. We adapt this approach to a classical setting, to obtain a method for transitioning efficiently between a simple tractable distribution and a challenging distribution of interest. We show that our approach is closely related to the proposals of [**albergo2024nets**, **vargas2023transport**, **arbel2021annealed**] among others, but in a Hamiltonian setting.

2 Related work

Sequential Monte Carlo In the context of statistics and Bayesian inference, several algorithms exist to deal with the problem of slow convergence to the target distribution. Most relevant to the present work is the application of Sequential Monte Carlo [**doucet2001introduction**] to sampling from static distributions [**del2006sequential**], by constructing a series of distributions with densities π_i where π_0 is the initial distribution and π_T is the target distribution. Here a series of particles are evolved according to a Markov kernel targeting the stationary distribution of π_i at step i, and at each step reweighting and resampling, so that the weighted particles are distributed according to p_i.

Several proposals have been made to replace the Markov kernel with one which more effectively transports the particles from π_i to π_{i+1}, including Controlled Sequential Monte Carlo [**heng2020controlled**] and Schrodinger Bridge SMC [**bernton2019schr**].

Dynamical transport A separate body of research approaches the problem from the perspective of stochastic optimal control [**fleming2012deterministic**] and optimal transport [**peyre2019computational**]. Here, one considers an ODE or SDE governing a variable of interest x, which gives rises to a PDE that governs the evolution of the corresponding density π. The goal is to design the ODE or SDE so that the density π evolves from a simple initial distribution to a chosen target distribution.

Among these, Controlled Monte Carlo Diffusions (CMCD [**vargas2023transport**]), the Liouville Flow Importance Sampler (LFIS [**tian2024liouville**]), Path Guided Particle Sampling (PGPS [**fan2024path**], the Path-Integral Sampler (PIS [**zhang2021path**]) and Non-equilibrium transport sampling (NETS [**albergo2024nets**]) all learn a drift term of a differential equation, parametrized by a neural network.

At the intersection of these two perspectives, several recent approaches [**albergo2024nets**, **vargas2023transport**] have proposed to use weighting schemes to correct for remaining bias, in the manner of SMC. These are inspired by Jarsynski's equality [**jarzynski1997nonequilibrium**] and more generally from the detailed fluctuation theorem [**crooks1998nonequilibrium**].

Statistical physics Finally, the problem of transitioning from one stationary distribution to another is a core topic of thermodynamics, and has practical implications for the preparation of physical systems. Adding a term to Hamiltonian dynamics to preserve stationarity through a series of intermediate distributions has been proposed several times in a quantum setting, as *engineered swift equilibration* [**martinez2016engineered**] and counterdiabatic driving [**demirplak2003adiabatic**], where the corresponding problem with a quickly changing Hamiltonian is failure to preserve eigenstates. The core ideas carry over to a classical setting [**patra2017shortcuts**], including with stochastic dynamics, often under the name *shortcuts to adiabaticity*, as well as *escorted free energy* (EFE) [**vaikuntanathan2011escorted**]. We recommend [**guery2019shortcuts**] for an overview. While the problem is closely related to the optimal transport and control perspectives above, the setting of Hamiltonian dynamics provides useful geometric insights, in particular that the learned correction to the Hamiltonian can be understood geometrically as a "gauge potential", or equivalently as a connection on a fiber bundle [**kolodrubetz2017geometry**].

The above papers focus on analytic solutions to the driving term for particular physical systems; meanwhile, [**sels2017minimizing**] proposes *learning* a driving term in a quantum context, but notes that the approach is transferable to a classical setting, which is a central inspiration for our approach.

Contributions Since Hamiltonian Monte Carlo is the gold standard approach for sampling from differentiable densities, it is natural that schemes for accelerating convergence to stationarity should focus on it. We emphasize this focus as the main ingredient by which our method differs from the recent approaches to sampling with learning discussed above.

Table 2 summarizes what we consider to be the most similar approaches to our own and the relevant differences. All of the listed methods take a population of particles, and evolve them, while also varying the distribution from an easy initial distribution to a target of interest. *Dynamics* refers to the way in which particles are updated at each step, *Weights* to whether a weighting scheme is used to correct the distribution of particles, *Learned term* to what, if any, term needs to be learned for the dynamics, and *Amortized* to whether the algorithm requires a phase before runtime in which training takes place (as opposed to any training happening during the evolution of the particles).

Method	Dynamics	Weights	Learned term	Amortized	Field
CSMC	Markov kernel	Yes	Kernel	Yes	Stats
EFE [**vaikuntanathan2011escorted**]	Hamiltonian	No	None	No	Phys/Chem
APTCF [**demirplak2003adiabatic**]	Hamiltonian	No	None	No	Chem
AIS [**neal2001annealed**]	MCMC	Yes	None	No	Stats
NETS [**albergo2024nets**]	Overdamped Langevin	Yes	Drift	Yes	ML
CMCD [**vargas2023transport**]	Langevin	No (?)	Drift	Yes	ML
AFT [**arbel2021annealed**]	NF + MCMC	Yes	NF Map	No	ML
LFIS [**tian2024liouville**]	ODE	No	Drift	No	ML
PGPS [**fan2024path**]	ODE	No	Drift	No	ML
DMT [**sun2024dynamical**]	PDE	No	Drift	No	ML/PDEs
PIS [**zhang2021path**]	SDE	Yes	Drift	Yes	ML/Control
CMC (ours)	Hamiltonian	No	Gauge field	No	ML/Phys

3 Counterdiabatic Monte Carlo

Hamiltonian dynamics Suppose we have a physical system given by a d dimensional vector $q \in \mathbb{R}^d$ d, such as the position of a particle. Hamilton's equations of motion are a 1st order ODE defined on $(q, p) \in \mathbb{R}^{2d}$, where p is a second variable called the momentum. The ODE is defined in terms of a real-valued *Hamiltonian* function $H(q, p)$.

$$\dot{q} = \nabla_p H(q,p) \tag{1}$$
$$\dot{p} = -\nabla_q H(q,p) \tag{2}$$

We refer to the solution to the ODE as the *flow* ϕ_H, so that $\phi_H(q_0, p_0, t)$ is the state of the system at time t given the initial state (q_0, p_0).

Poisson bracket If f is also a real valued function of (q,p), then the time derivative $\partial_t f$ can be written as the *Poisson bracket* $\partial_t f(q,p) = \{f, H\}(q,p) := \nabla_q f \cdot \nabla_p H - \nabla_p f \cdot \nabla_q H$.

We note that the Poisson bracket is antisymmetric, i.e. $\{A, B\} = -\{B, A\}$, and that $\int C(q,p)\{A,B\}(q,p)dqdp = \int B(q,p)\{C,A\}(q,p)dqdp$.

Canonical distribution The *canonical distribution* defined as $\rho_H(q,p) \propto e^{-H(q,p)}$ is a stationary distribution of Hamiltonian dynamics. Indeed, the continuity equation $\partial_t \rho_H = \{H, \rho_H\}$ implies that $\partial_t \rho_H = 0$, by an above property of the Poisson bracket. We denote by ϕ^*_H the flow of the differential equation, so that $\phi^*_H(t')(\rho_H) = \rho_H$.

Hamiltonian Monte Carlo The principle of Hamiltonian Monte Carlo is to use a discretization of Hamiltonian dynamics to produce a kernel which has a stationary distribution ρ_H for which the marginal on q, i.e. $\pi(q) \propto \int \rho_H(q,p)dp$, is the distribution of interest. This can be achieved by choosing $H = -\nabla \log \pi(q) + \frac{1}{2}|p|^2$, which only requires knowing the density up to the normalization constant.

The Metropolis-Hastings algorithm [**chib1995understanding**] is used to correct for bias coming from the discretization of the Hamiltonian dynamics, and momentum can be refreshed every n steps, or partially refreshed every step, in order to ensure ergodicity.

3.1 Counterdiabatic driving

When H is time dependent, for example through a parameter λ, we can consider the time dependent flow $\phi_{H(\lambda(t))}$. Crucially, we note that $\partial_t \rho_{H(\lambda(t))} \neq 0$, so that $\phi^*_H(t')\rho_{H(\lambda(t))} \neq \rho_{H(\lambda(t+t'))}$. In other words, the flow of the time-varying Hamiltonian does not preserve the time-varying distribution $\rho_{H(\lambda(t))}$. If it did, we could simply run Hamiltonian dynamics with $H(\lambda(t))$ and obtain at time T the intended target perfectly. To see why it fails, we proceed as follows:

$$\partial_t \rho_{H(\lambda(t))} = \{H, \rho_{H(\lambda(t))}\} + \dot{\lambda}\partial_\lambda \rho_{H(\lambda(t))}$$

$$= 0 - \dot{\lambda}\partial_\lambda H \rho_{H(\lambda(t))} - \dot{\lambda}\frac{\partial \log Z}{\partial \lambda}\rho_{H(\lambda(t))} = (-\partial_\lambda H + \langle \partial_\lambda H \rangle_t)\dot{\lambda}\rho_{H(\lambda(t))}$$

[**demirplak2003adiabatic**], [**berry2009transitionless**] and [**vaikuntanathan2011escorted**, **guery2023driving**] note[1] that one can offset the lag by introducing a term A, known as the *counterdiabatic gauge potential*, with the property that $\{A, H\} = \partial_\lambda H - \langle \partial_\lambda H \rangle_t$. Then we would have:

$$\partial_\lambda \rho_{H(\lambda(t))} = \{H, A\}\rho_{H(\lambda(t))} = \{\rho_{H(\lambda(t))}, -A\} = \{A, \rho_{H(\lambda(t))}\}$$

This can be interpreted as the evolution of the distribution that would be generated by a Hamiltonian A. This demonstrates a remarkable result: if we choose as our Hamiltonian $H_c := H + \dot{\lambda}A$, we have that $\phi^*_{H+\dot{\lambda}A}(t')\rho_{H(\lambda(t))} = \rho_{H(\lambda(t+t'))}$. In other words, by using H_c as our Hamiltonian, we obtain perfect transport to each ρ_{H_t}, including ρ_{H_T}.

[1] Here, the focus is on quantum systems, but by the correspondence between Poisson brackets and commutators, the central insight may be translated to a classical setting, as explained in [].

Learning the counterdiabatic gauge potential [**sels2017minimizing**] demonstrates that one can variationally approximate A in a computationally practical way. In classical terms, the key idea is that we minimize $\mathbb{E}_{z\sim\rho_{H(\lambda(t))}}[|G(z)|^2]$ at each time point t, where

$$G(z) = \{A, H\}(z) - \partial_\lambda H(z) + \mathbb{E}[\partial_\lambda H(z)]$$

Multiplying out, we find $\mathbb{E}[|G(z)|^2] = \mathbb{E}[\{A,H\}^2 + (\partial_\lambda H)^2 + \mathbb{E}[\partial_\lambda H]^2 - 2\{A,H\}(\partial_\lambda H) + 2\{A,H\}\mathbb{E}[\partial_\lambda H] - 2(\partial_\lambda H)\mathbb{E}[\partial_\lambda H]] = \mathbb{E}[\{A,H\}^2 + (\partial_\lambda H)^2 - 2\{A,H\}(\partial_\lambda H) + 2\{A,H\}\mathbb{E}[\partial_\lambda H] - \mathbb{E}[\partial_\lambda H]^2]$

We now note that when $\partial_\lambda H = \{A, H\}$, we have that $\langle \partial_\lambda H\rangle = \langle\{A,H\}\rangle = \int \rho\{A,H\}dz = \int A\{\rho, H\} = 0$, where the penultimate step follows by an above property of the Poisson bracket. This means that $\mathbb{E}[G(z)] = \mathbb{E}[\{A,H\}^2 + (\partial_\lambda H)^2 - 2\{A,H\}(\partial_\lambda H)] = \mathbb{E}[|\{A,H\}(q,p) - \partial_\lambda H(q,p)|^2] := \mathcal{L}_H(A)$.

This means that we can minimize the loss $\mathcal{L}_H(A)$ instead, which is simple to calculate.

3.2 Weighting

When A is learned perfectly, and the discretization step size is taken to zero, transport is perfect. However, in practice, we must use a finite step size, and we must approximate A. This introduces discrepancy between the distribution of the particles at time t and $\rho_{H(t)}$, which we can correct for with weights if necessary.

When a weighting scheme is used, our proposed algorithm is a special case of the general framework of Sequential Monte Carlo samplers [**doucet2001sequential**], as shown in appendix A.

Crucially, the estimate of the expectation $\mathbb{E}[R(A, H)]$ can be calculated using the weighted particles, which improves the learning of A.

$$dW = \frac{\partial H}{\partial t} - \nabla H \cdot \nabla A + \nabla^2 A$$

which defines $W[z(t)] = \int_0^T dW$

4 Algorithm design

The fact that we can learn an approximation of A suggests an algorithm for sampling. We first choose a path of densities $\pi(q; \lambda(t))$ where $\pi(q; \lambda(0))$ is chosen to be straightforward to sample from, and where $\pi(q; \lambda(T))$ is the distribution of interest. This defines a time varying Hamiltonian $H(p, q; \lambda(t)) = -\nabla \log \pi(q; \lambda(t)) + \frac{1}{2}|p|^2$.

We then initialize a population of particles in $\pi(\cdot; \lambda(0))$, distributed according to $\rho_{H(\lambda(0))}$. At each time t, discretized in steps of length ϵ, we minimize the *expectation* of R, under the empirical distribution of the current population, to obtain an estimate of A (Algorithm 1). We use $H(\cdot; \lambda(t)) + \dot{\lambda}A(\cdot; \lambda(t))$ as our Hamiltonian at a given time t. Further, we have the option of computing weights according to (**??**) after each time step, which correct for the bias introduced by the numerical discretization of the dynamics and the approximation of A. This is summarized in Algorithm 2.

This can be understood as a special case of the general framework of Sequential Monte Carlo samplers [**doucet2001sequential**], as shown in appendix TODO.

Numerical Integration A key complication is that A is not separatable into a term depending on p and a term depending on q. This means that the Velocity Verlet integrator is not symplectic.

Choice of step size As in standard HMC, the choice of ϵ is a trade-off between the accuracy of the dynamics and the efficiency of the algorithm. We propose to tune step size by monitoring the error in H; in the limit of small step size and perfect A, H should be preserved exactly.

consider fisher score

Algorithm 1 Fit adiabatic gauge potential A_ϕ at a given λ

Require: Real-valued function $A_\phi(q, p; \lambda)$, optimizer (e.g. Adam) , schedule λ_i, population of particles $\{q_i, p_i\}_{i=1}^N$, number of iterations T.

for $t = 1, \dots, T$ **do**
  Compute $l \leftarrow \frac{1}{N}\sum_{i=1}^N \{A_\phi, H_\lambda\}(q_i, p_i) - \partial_\lambda U(q_i; \lambda)$
  Update ϕ using optimizer with loss l
end for
return ϕ

Algorithm 2 Counterdiabatic HMC (CHMC) with population and weighting

Require: Population $\{q_i^{(0)}\}_{i=1}^N$ from initial distribution ; potential $U(q; \lambda)$ and gradients ; step size ε ; schedule $\{\lambda_k\}_{k=1}^L$ with $\lambda_1 = 0$ (easy) and $\lambda_L = 1$ (target) ; learned $A_\phi(q, p; \lambda)$ and its partials $\nabla_q A_\phi, \nabla_p A_\phi$.

Initialize weights $w_i^{(0)} = 1/N$ for all i
for $k = 1$ to $L - 1$ **do**
  for $i = 1$ to N **do**
    Sample $p_i \sim \mathcal{N}(0, I)$
    Set $(q, p) \leftarrow (q_i^{(k-1)}, p_i)$; set $W_i \leftarrow 0$
    Half-kick (potential & CD-q force):

$$p \leftarrow p - \tfrac{\varepsilon}{2}\Big(\nabla_q U(q; \lambda_k) + \dot{\lambda}_k \, \nabla_q A_\phi(q, p; \lambda_k)\Big)$$

    Drift (kinetic & CD-p flow):

$$q \leftarrow q + \varepsilon\Big(p + \dot{\lambda}_k \, \nabla_p A_\phi(q, p; \lambda_k)\Big)$$

    Protocol work increment (discrete $\partial_t H^{\mathrm{CD}}$):

$$W_i \leftarrow W_i + \Big(U(q; \lambda_{k+1}) - U(q; \lambda_k)\Big) + \Big(\dot{\lambda}_{k+1} A_\phi(q, p; \lambda_{k+1}) - \dot{\lambda}_k A_\phi(q, p; \lambda_k)\Big)$$

    Half-kick (with updated waypoint):

$$p \leftarrow p - \tfrac{\varepsilon}{2}\Big(\nabla_q U(q; \lambda_{k+1}) + \dot{\lambda}_{k+1} \, \nabla_q A_\phi(q, p; \lambda_{k+1})\Big)$$

    Update particle: $q_i^{(k)} \leftarrow q$
    Update weight: $w_i^{(k)} \leftarrow w_i^{(k-1)} \exp(-W_i)$
  end for
  Normalize weights: $w_i^{(k)} \leftarrow w_i^{(k)} / \sum_j w_j^{(k)}$
  Optionally resample if effective sample size is low
end for
return weighted population $\{(q_i^{(L)}, w_i^{(L)})\}_{i=1}^N$

5 Evaluation

The goal of CHMC is to reduce the wall-clock time required to sample from challenging distributions. We consider the regime where the gradient of the log density is costly, so that our central focus is on the number of calls to this gradient function required for an *effective sample size* of 100 samples.

We compare to Sequential Monte Carlo, as well as the results reported in NETS TODO

We use the models proposed in the NETS paper for evaluation (todo: get michael or eric to share them).

6 Conclusion

A Counterdiabatic HMC as an SMC sampler

Given an initial distribution $\mu \in \mathcal{P}(\mathsf{X})$ and a sequence of Markov kernels $M_t \in \mathcal{M}(\mathsf{X})$ for $t = 1, \ldots, T$, the *proposal* (or reference) path measure $\mathbb{Q}$ is

$$\mathbb{Q}(\mathrm{d}x_{0:T}) = \mu(\mathrm{d}x_0) \prod_{t=1}^{T} M_t(x_{t-1}, \mathrm{d}x_t). \tag{3}$$

Importance sampling on path space (Feynman–Kac). Let $G_0 : \mathsf{X} \to (0, \infty)$ and $G_t : \mathsf{X} \times \mathsf{X} \to (0, \infty)$ for $t \geq 1$ be strictly positive *potentials*. The *target* path measure $\mathbb{P}$ is defined by reweighting $\mathbb{Q}$:

$$\mathbb{P}(\mathrm{d}x_{0:T}) = \frac{1}{Z} G_0(x_0) \left[\prod_{t=1}^{T} G_t(x_{t-1}, x_t) \right] \mathbb{Q}(\mathrm{d}x_{0:T}), \tag{4}$$

where the normalizing constant is $Z = \mathbb{E}_{\mathbb{Q}}\Big[G_0(X_0) \prod_{t=1}^{T} G_t(X_{t-1}, X_t)\Big]$. The resulting unnormalized *marginal* measures $\gamma_t \in \mathcal{S}(\mathsf{X})$ and their normalized counterparts $\eta_t \in \mathcal{P}(\mathsf{X})$ are

$$\gamma_t(\varphi) = \mathbb{E}_{\mathbb{Q}}\left[\varphi(X_t)\, G_0(X_0) \prod_{s=1}^{t} G_s(X_{s-1}, X_s)\right], \qquad Z_t := \gamma_t(1), \tag{5}$$

$$\eta_t(\varphi) = \gamma_t(\varphi)/Z_t, \qquad t = 0, \ldots, T. \tag{6}$$

Eqs. (3)–(5) constitute the standard Feynman–Kac framework: importance sampling is performed on the *path* measure, while particle approximations target the time marginals $(\eta_t)_{t=0:T}$.

One can then sample from todo by initializing a population of particles from π_0 and then propagating them forward in time using the kernel M_t, keeping track of the weights at each step and optionally resampling to mitigate weight degeneracy.

Sequential Monte Carlo samplers [**doucet2001sequential**] show how the general framework of Sequential Monte Carlo as summarized above can be applied to the problem of sampling from a distribution π on $\mathbb{R}^d$, by artificially constructing a path of distributions π_t with π_0 a simple distribution and $\pi_T = \pi$ the target distribution.

The central idea is that a forward path can be constructed by choosing a time dependent Markov kernel M_t

Q(M, pi0)

and that this can be renormalized to sample from a

Q(L, piT)

This requires $G = ...$

SMC with physical dynamics: nonequilibrium fluctuation theorems In the case where M and L are given by forward time and backward time Hamiltonian dynamics (or overdamped or underdamped Langevin dynamics), the ratio of $M(x_t|x_{t-1})$ to $L(x_{t-1}|x_t)$ can be calculated using the non-equilibrium fluctuation theorem derived by [**crooks1998nonequilibrium**] and extended to more generality [**todo**].

First, recall that

$$\frac{dH}{dt} = \frac{\partial H}{\partial t} + \nabla H \cdot \frac{d\mathbf{x}}{dt} := dW + dQ$$

In physics terms, dW and dQ are the work done by the system and the heat absorbed by the system, respectively, and mathematically, should be viewed as one-forms that are not exact, i.e. are not the differential of a function. By contrast, their sum dH is an exact one-form, namely the differential of the Hamiltonian.

In particular, this theorem states that

$$\frac{\pi(z_{t-1})k(z_t|z_{t-1})}{\pi(z_t)k(z_{t-1}|z_t)} = \exp^{W-\Delta F}$$

where W is the integral of $dw = \frac{\partial H}{\partial t}$ over the trajectory, and $\Delta F = -\log \frac{Z_t}{Z_{t-1}}$, for $Z_t := \int \pi_{H(t)}(z)dz$.

The consequence is that we can compute

TODO: define k

$$\frac{k(z_t|z_{t-1})}{k(z_{t-1}|z_t)} = e^W \frac{Z_t}{Z_{t-1}} \frac{Z_{t-1}}{Z_t} e^{-\Delta H} = e^{W-\Delta H} = e^{-Q}$$

Since Hamiltonian dynamics are measure-preserving, i.e. $Q = 0$, even when numerically simulated with a leapfrog integrator, we see immediately that when using an unadjusted HMC kernel in the SMC sampling algorithm, the correct weight is given by $e^{-\Delta H}$, in agreement with Appendix B of [**dai2022invitation**].

The more interesting case concerns a kernel k that arises from a counterdiabatic Hamiltonian H_c. In this case, [**zhong2024time**] derives an analogous theorem, but in terms of a different work, which we refer to as W_c, obtained by integrating $dw_c = todo$ over the trajectory, where

$$dw_c = (\frac{\partial H_c}{\partial t} - \nabla H \cdot v_A + \nabla \cdot v_A)dt$$

where v_A is the drift induced by A, i.e. $v_A = (\nabla_q A, -\nabla_p A)$. We then see that the third term vanishes, since Hamiltonian flow is dissipationless. In this case, we see that the physical intepretation of heat and work is preserved, in the sense that

$$\frac{dH}{dt} = \frac{\partial H}{\partial t} + \nabla H \cdot v + \nabla H \cdot v_H$$

where v_H is the drift induced by H, i.e. $v_H = (\nabla_q H, -\nabla_p H)$. This means that $dw_c + dq = dH$, where $\nabla H \cdot v_H$ is defined as before.

By definition of

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